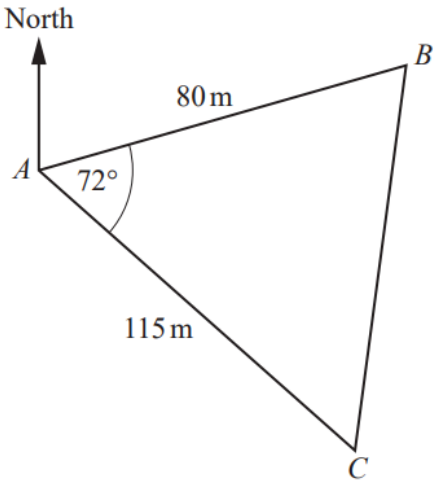
TRIGONOMETRY

Topical Practice • Paper 4

Past Paper Topical Questions (Paper 4)

1. Q(7) 0580/41/M/J/20



NOT TO SCALE

The diagram shows the positions of three points A, B and C in a field.

(a) Show that BC is 118.1 m, correct to 1 decimal place.

[3]

(b) Calculate angle ABC.

Angle ABC = ... [3]

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(d) Mitchell takes 35 seconds to run from A to C.

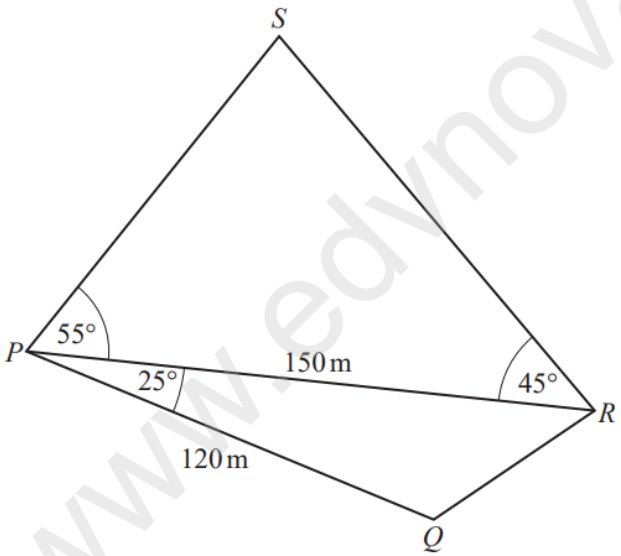
Calculate his average running speed in kilometres per hour.

..... km/h [3]

(e) Calculate the shortest distance from point B to AC.

..... m [3]

2. Q(4) 0580/42/M/J/20



NOT TO SCALE

The diagram shows two triangles.

(a) Calculate QR.

QR = ... m [3]

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(b) Calculate RS.

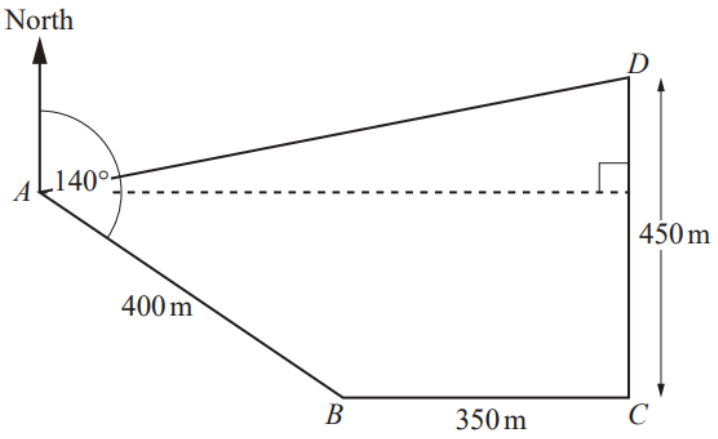
$RS =$ ..... m [4]

(c) Calculate the total area of the two triangles.

..... m^2 [3]

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3. Q(5b) 0580/42/M/J/20



NOT TO SCALE

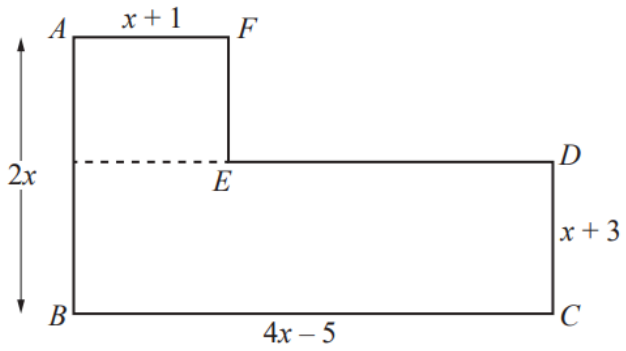
The diagram shows a field $ABCD$.
The bearing of B from A is 140°.
C is due east of B and D is due north of C.
$AB = 400$ m, $BC = 350$ m and $CD = 450$ m.

(b) Calculate the distance from D to A.

..... m [6]

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4. All the lengths in this question are in centimetres. Q(5d) 0580/43/M/J/20



NOT TO SCALE

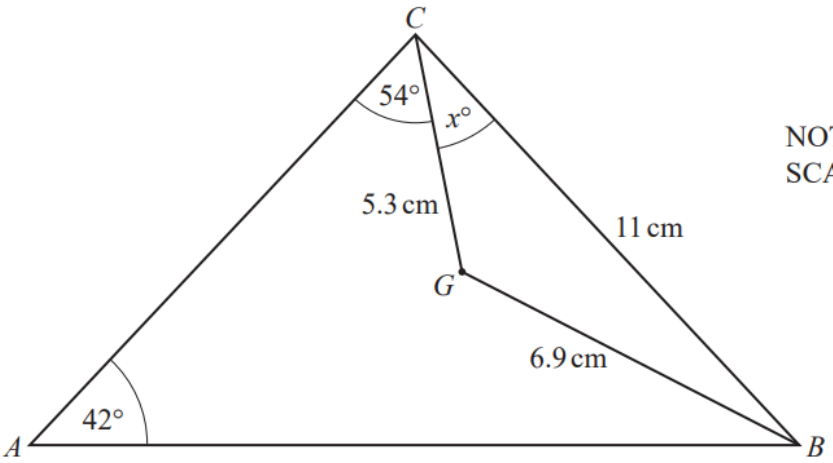
The diagram shows a shape *ABCDEF* made from two rectangles.
The total area of the shape is 342 cm^2.

(d) Calculate angle *DBC*.

Angle *DBC* = ... [2]

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5. (a) Q(6a) 0580/43/M/J/20



The diagram shows triangle ABC with point G inside.
$CB = 11$ cm, $CG = 5.3$ cm and $BG = 6.9$ cm.
Angle $CAB = 42°$ and angle $ACG = 54°$.

(i) Calculate the value of x.

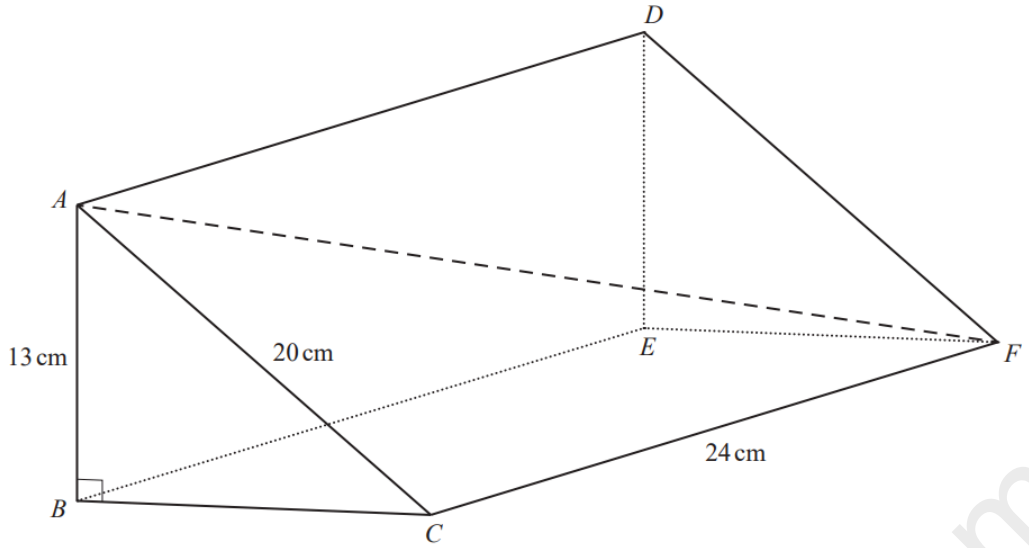
$x =$ ..... [4]

(ii) Calculate AC.

$AC =$ ..... cm [4]

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6. Q(9) 0580/41/M/J/21



NOT TO SCALE

The diagram shows a prism, $ABCDEF$.
$AB = 13$ cm, $AC = 20$ cm, $CF = 24$ cm and angle $ABC = 90°$.

(a) Calculate the total surface area of the prism.

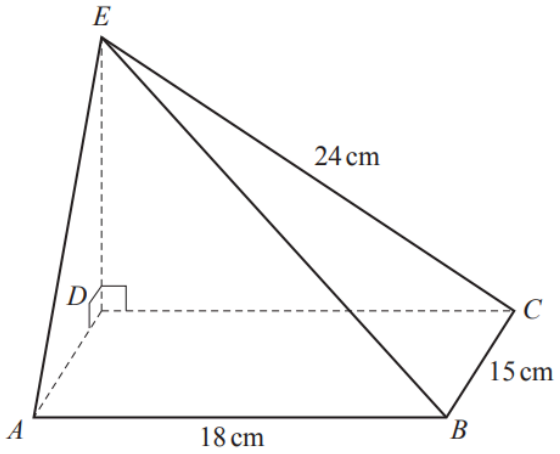
..... cm^2 [6]

(c) Calculate the angle that AF makes with the base $BCFE$.

..... [4]

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7. (b) Q(8b) 0580/42/M/J/21



NOT TO SCALE

The diagram shows a solid pyramid with a rectangular base $ABCD$.
E is vertically above D.
Angle EDC = angle EDA = 90°.
AB = 18 cm, BC = 15 cm and EC = 24 cm.

(i) The pyramid is made of wood and has a mass of 800 g.

Calculate the density of the wood.
Give the units of your answer.

[The volume, V, of a pyramid is $V = \frac{1}{3} \times$ area of base $\times$ height.]
[Density = mass ÷ volume]

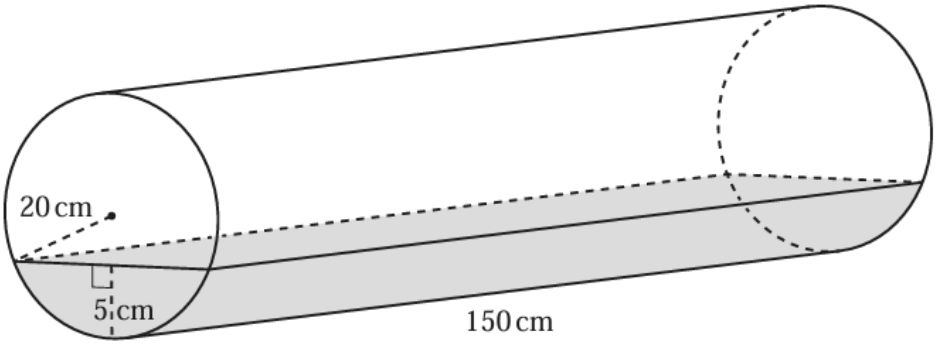
.... [5]

(ii) Calculate the angle between BE and the base of the pyramid.

..... [4]

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8. Q(8c) 0580/43/M/J/21



NOT TO SCALE

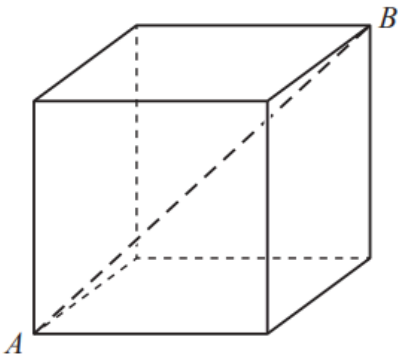
The diagram shows a cylinder of length 150 cm on horizontal ground.
The cylinder has radius 20 cm.
The cylinder contains water to a depth of 5 cm, as shown in the diagram.

Calculate the volume of water in the cylinder.
Give your answer in litres.

..... litres [7]

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9. (b) Q(9b) 0580/43/M/J/21



NOT TO SCALE

The diagram shows a cube.
The length of the diagonal AB is 8.5 cm.

(i) Calculate the length of an edge of the cube.

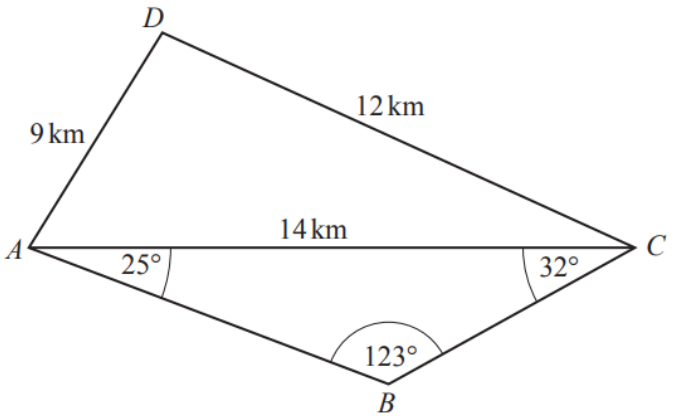
..... cm [3]

(ii) Calculate the angle between AB and the base of the cube.

..... [3]

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10. Q(7) 0580/41/M/J/22



NOT TO SCALE

(a) Calculate angle ACD.

Angle ACD = ... [4]

(b) Show that $BC = 7.05\,\text{km}$, correct to 2 decimal places.

[3]

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(c) Calculate the shortest distance from B to AC.

..... km [3]

(d) Calculate the length of the straight line BD.

BD = ... km [4]

(e) C is due east of A.

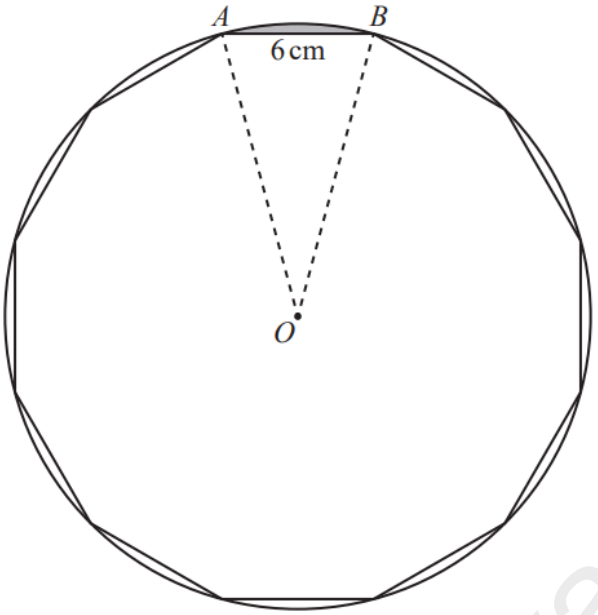
Find the bearing of D from C.

..... [2]

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11. (b) The polygon is enclosed by a circle, centre O, so that each vertex touches the circumference of the circle.

Q(4b) 0580/43/M/J/22



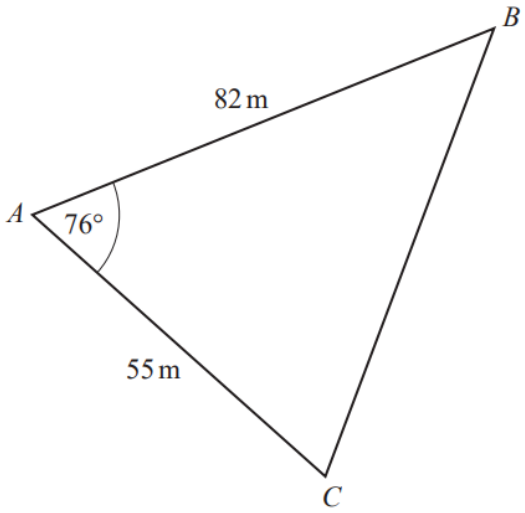
NOT TO SCALE

(i) Show that the radius, AO, of the circle is 11.6 cm, correct to 1 decimal place.

[3]

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12. Q(7) 0580/43/M/J/22



NOT TO SCALE

The diagram shows a field ABC.

(a) Calculate BC.

$BC =$ ..... m [3]

(b) Calculate angle ACB.

Angle $ACB =$ ..... [3]

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(c) A gate, G, lies on AB at the shortest distance from C.

Calculate AG.

$AG =$ ..... m [3]

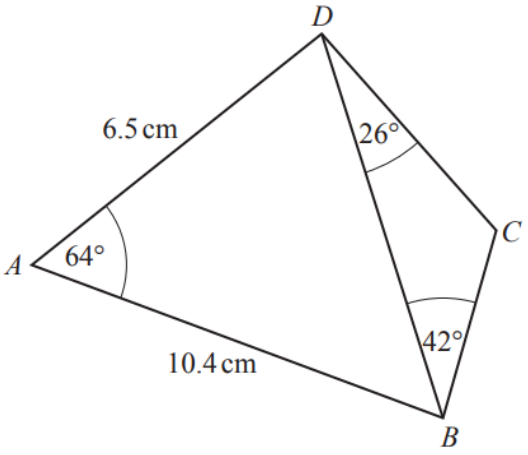
(d) A different triangular field PQR has the same area as ABC.
$PQ = 90\,\text{m}$ and $QR = 60\,\text{m}$.

Work out the two possible values of angle PQR.

Angle $PQR =$ ..... or ..... [5]

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13. Q(6) 0580/42/M/J/24



NOT TO SCALE

$ABCD$ is a quadrilateral with $AB = 10.4$ cm and $AD = 6.5$ cm.
Angle $DAB = 64°$, angle $BDC = 26°$ and angle $DBC = 42°$.

(a) Show that $BD = 9.55$ cm, correct to 2 decimal places.

[3]

(b) (i) Show that angle $BCD = 112°$.

[1]

(ii) Calculate CD.

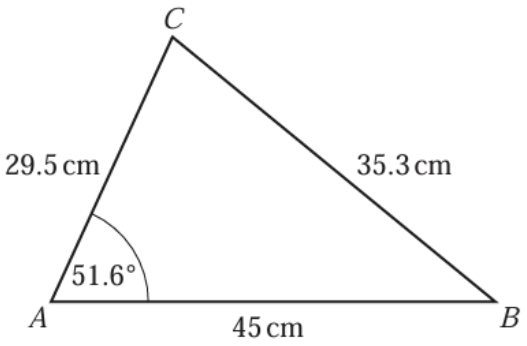
$CD =$ ..... [3]

(c) Find the shortest distance from D to AB.

..... cm [3]

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14. (a) Q(4) 0580/42/F/M/22



NOT TO SCALE

In triangle ABC, $AB = 45$ cm, $AC = 29.5$ cm, $BC = 35.3$ cm and angle $CAB = 51.6°$.

(i) Calculate angle ABC.

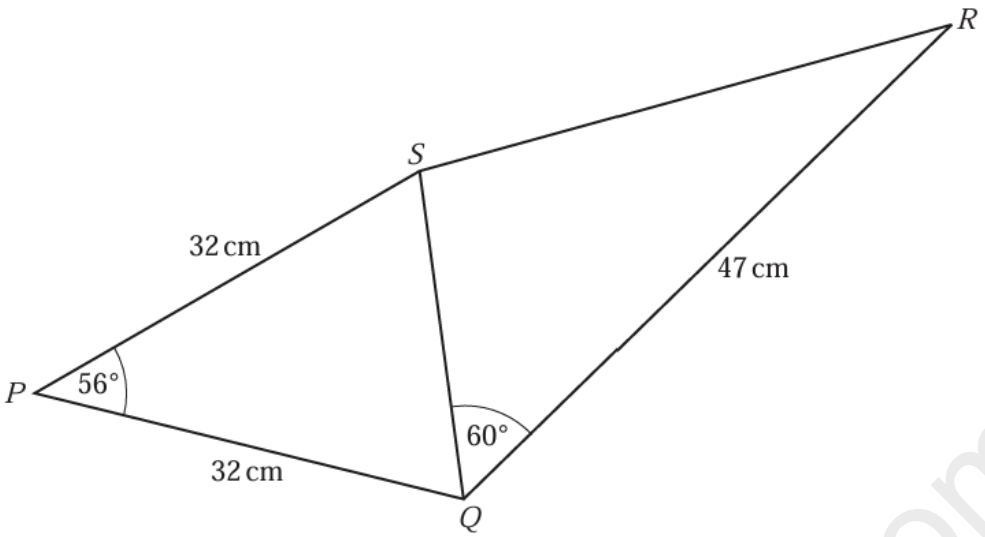
Angle $ABC =$ ..... [3]

(ii) Calculate the area of triangle ABC.

..... cm^2 [2]

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(b)



NOT TO SCALE

The diagram shows a quadrilateral $PQRS$ formed from two triangles, PQS and QRS.
Triangle PQS is isosceles, with $PQ = PS = 32$ cm and angle $SPQ = 56°$.
$QR = 47$ cm and angle $SQR = 60°$.

(i) Calculate SR.

$SR =$ ..... cm [4]

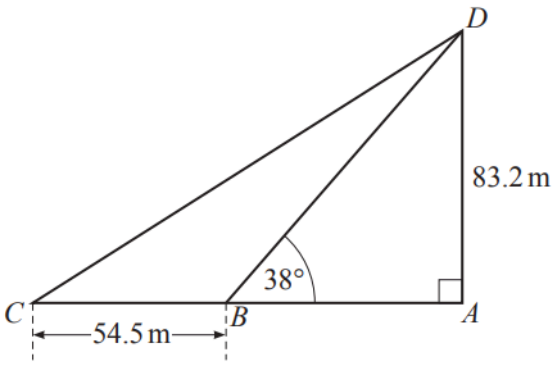
(ii) Calculate the shortest distance from P to SQ.

..... cm [3]

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15. (a)

Q(5) 0580/41/O/N/23



NOT TO SCALE

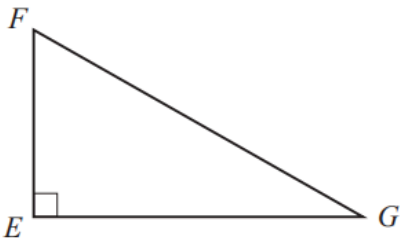
ACD is a right-angled triangle.
B is on AC and $BC = 54.5$ m.
$AD = 83.2$ m and angle $ABD = 38°$.

Calculate angle ACD.

Angle ACD = ... [5]

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(b)



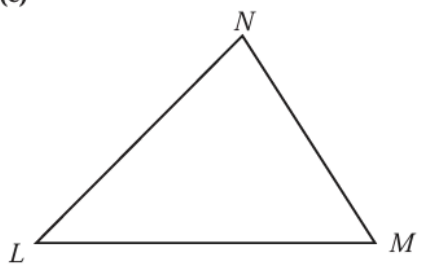
EFG is a right-angled triangle.
A circle can be drawn that passes through the three vertices of the triangle.

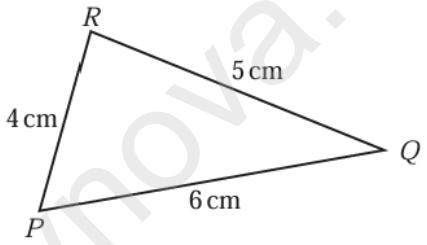
On the diagram, mark the position of the centre of the circle with a cross.
Explain how you decide.

.....

..... [2]

(c)





NOT TO SCALE

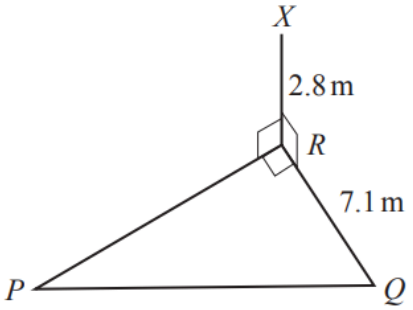
In triangle *LMN*, the ratio angle *L* : angle *M* : angle *N* = 4 : 5 : 6.
In triangle *PQR*, *PQ* = 6 cm, *PR* = 4 cm and *QR* = 5 cm.

Calculate the difference between the largest angle in triangle *PQR* and the largest angle in triangle *LMN*.

..... [7]

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16. (a) Q(7) 0580/42/O/N/23



NOT TO SCALE

The diagram shows a right-angled triangle PQR on horizontal ground.
X is vertically above R and the angle of elevation of X from P is 21°.
$XR = 2.8$ m and $RQ = 7.1$ m.

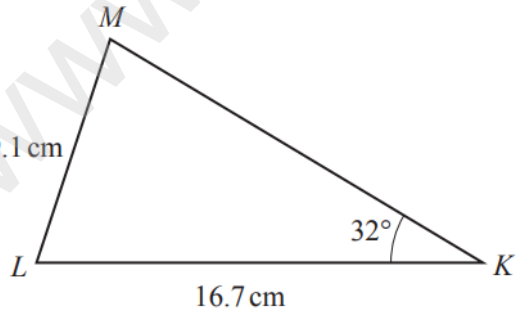
(i) Calculate the angle of elevation of X from Q.

..... [2]

(ii) Calculate PQ.

..... m [3]

(b)



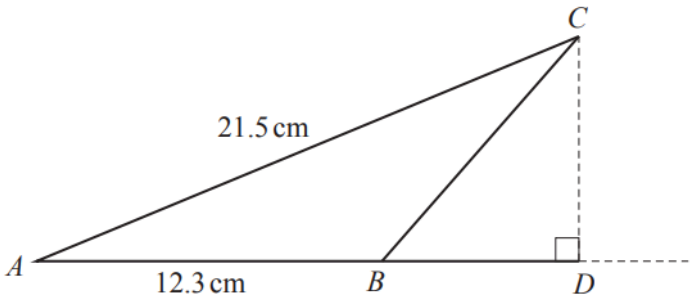
NOT TO SCALE

Calculate the acute angle KML.

Angle KML = ... [3]

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(c)



NOT TO SCALE

The area of triangle ABC is $62.89\,\text{cm}^2$.

(i) Show that angle $BAC = 28.4°$, correct to 1 decimal place.

[2]

(ii) Calculate BC.

..... cm [3]

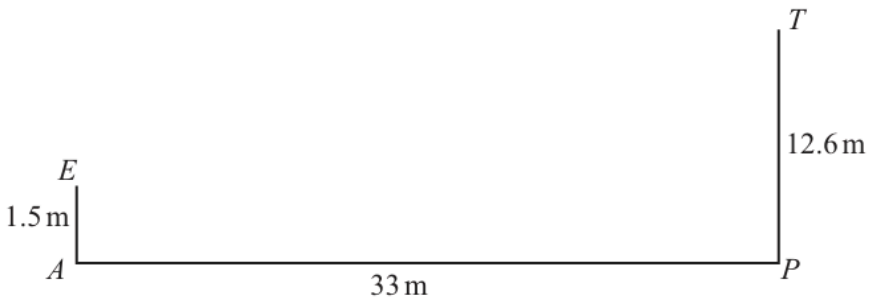
(iii) AB is extended to a point D such that angle $BDC = 90°$.

Calculate BD.

..... cm [3]

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17. Q(13) 0580/41/O/N/25



NOT TO SCALE

The diagram shows two vertical poles, AE and PT, standing on horizontal ground, AP.

Calculate the angle of elevation of the point T from the point E.

..... [3]

18. In triangle STU, $ST = 8$ cm, $SU = 9$ cm and angle $TSU = 50°$. Q(21) 0580/41/O/N/25

Calculate the area of triangle STU.

..... cm^2 [2]

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